

IEEE 802.11

Basics:

1. What is the difference between infrastructure and ad-hoc mode?

the main difference between infrastructure and ad-hoc mode is the **presence of a central access point in infrastructure mode for managing communication in the network**, while **ad-hoc mode allows devices to communicate directly with each other without the need for a central coordination point**.

2. What is the purpose of the distribution system?

Connectivity, Traffic Routing, Load Balancing, Scalability, Redundancy and Resilience, Security

3. Which protocol layers does the 802.11 standard comprise?

Physical Layer
Medium Access Control (MAC) Layer
Logical Link Control

Internet Protocol (IP)
Transport Layer
Application Layer

4. What are the main differences between the different PHY variants?

The main differences between PHY variants in the 802.11 standard stem from differences in frequency band, channel width, modulation and coding schemes, data rates, compatibility, spatial streams, and channel access mechanisms.

These differences allow PHY variants to adapt to different deployment scenarios, environments, and user requirements while providing flexibility, performance, and interoperability in Wi-Fi networks

MAC:

1) Exponential backoff: what does it mean? What is its purpose?

Exponential backoff involves increasing the waiting time between retransmissions after a collision occurs. When a device detects a collision during transmission it stops transmitting and waits for a random amount of time before attempting to retransmit the data.

Purpose:

The purpose of exponential backoff is to reduce the likelihood of collisions and improve the efficiency of channel utilization in shared communication channels.

2) Describe DCF and PCF. What are the differences, pros, and cons of each approach?

DCF (Distributed Coordination Function) and PCF (Point Coordination Function) are two coordination methods defined in the IEEE 802.11 standard for Wi-Fi networks.

DCF Purpose:

It allows devices to contend for access to the wireless medium by using Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

DCF Operation: Devices listen to the channel to detect if it is busy before transmitting data. If the channel is idle, the device waits for a random backoff period before attempting to transmit. If the channel is busy, the device defers its transmission until the channel becomes idle

DCF Pros:

Simple and Decentralized , Flexibility

DCF Cons:

Vulnerable to Collisions, Lack of prioritization

PCF Purpose:

PCF is an optional coordination method in Wi-Fi networks that provides centralized control over channel access. It allows an access point (AP) to coordinate access to the medium and allocate time slots for device transmissions

PCF Operation:

In PCF, the AP controls the channel access and periodically sends beacon frames to announce the start of contention-free periods (CFPs). During CFPs, the AP grants devices permission to transmit by allocating time slots for their transmissions. Devices can transmit data without contention during their allocated time slots, ensuring deterministic access to the medium.

PCF Pros:

Deterministic access, Prioritization

PCF Cons:

Complexity , Limited scalability

Summary:

DCF and PCF offer different approaches to channel access and coordination in Wi-Fi networks. DCF provides decentralized contention-based access, while PCF offers centralized control and deterministic access through time-division multiplexing. The choice between DCF and PCF depends on factors such as network size, traffic patterns, QoS requirements, and scalability considerations

3) DCF: what is the purpose of different interframe spaces?

In DCF different interframe spaces (IFS) serve the purpose of prioritizing access to the medium based on the type of frame being transmitted and ensuring fairness and efficiency in channel access.

The purpose of having different interframe spaces in DCF is to provide a mechanism for prioritizing access to the medium based on the urgency and importance of the frames being transmitted. Frames requiring immediate attention, such as ACK frames and response frames, are given the highest priority with the shortest interframe space (SIFS), ensuring minimal delay and efficient handling

4) Assume a node in DCF mode with RTS/CTS activated wants to transmit multiple segments of data: are multiple RTS/CTS cycles required for this?

No, multiple Request to Send/Clear to Send (RTS/CTS) cycles are not required for transmitting multiple segments of data in DCF (Distributed Coordination Function) mode with RTS/CTS activated.

5) For which messages does the SIFS apply? For what reason?

The Short Interframe Space (SIFS) applies primarily to high-priority messages in the IEEE 802.11 wireless LAN protocol.

Reason:

SIFS is necessary for these critical control and management frames to ensure timely delivery and response, thereby enhancing the overall performance and reliability of the wireless LAN protocol.

6) What is the purpose of the NAV?

The Network Allocation Vector (NAV) is a mechanism used in IEEE 802.11 wireless networks to coordinate access to the medium and prevent collisions.

Its purpose is to inform other stations in the network about the duration of ongoing transmissions, including the transmission of frames such as Data, Management, and Control frames.

The main purposes of NAV are:

- Collision Avoiding
- Virtual Carrier sensing
- Fairness and Efficiency

7) How does DCF handle situations with high loads? What are the strategies?

Some of the key strategies used by DCF in high-load scenarios include:

- Binary Exponential Backoff
- Contention Window Adjustment
- Frame aggregation
- Prioritization Mechanisms
- Back off free periods

MAC Management:

1) How does beaconing work in ad-hoc mode?

In ad-hoc mode, where devices communicate directly with each other without the need for a central access point (AP), beaconing works slightly differently compared to infrastructure mode where APs periodically broadcast beacon frames to announce their presence and network parameters. In ad-hoc mode, devices themselves take on the responsibility of beaconing to facilitate communication and coordination within the ad-hoc network.

beaconing in ad-hoc mode enables devices to discover each other, exchange network information, coordinate communication, and synchronize their operations within the self-configured ad-hoc network. While there is no central AP to manage beacon transmissions, each device autonomously beacon frames to facilitate network operation and communication.

2) How are collisions among beacons avoided in ad-hoc mode?

Collisions among beacons in ad-hoc mode are avoided by using a random backoff mechanism before transmitting beacon frames. Each device in the ad-hoc network selects a random backoff interval before transmitting its beacon frame. This randomization helps distribute beacon transmissions evenly across devices, reducing the likelihood of collisions and ensuring efficient beaconing in the network. Additionally, devices monitor the medium before transmitting their beacon frames to detect if it is busy, further reducing the risk of collisions.

3) What's the idea behind energy saving in 802.11?

The idea behind energy saving in 802.11 networks is to reduce power consumption in wireless devices by allowing them to enter low-power sleep modes when idle or not actively transmitting or receiving data. This is achieved through mechanisms such as Power Save Mode (PSM) and the use of periodic beacon frames to announce upcoming traffic, allowing devices to wake up selectively to receive data only when necessary, thereby conserving energy.

4) Does power saving work in ad-hoc mode as well?

Yes, power saving mechanisms such as Power Save Mode (PSM) can work in ad-hoc mode as well. Devices in ad-hoc mode can enter low-power sleep modes when idle or not actively transmitting or receiving data to conserve energy. Additionally, devices can use techniques such as periodic beacon frames to announce upcoming traffic, allowing them to wake up selectively to receive data only when necessary, even in ad-hoc networks.

5) What information does a TIM contain? What a DTIM?

A TIM (Traffic Indication Map) contains information about which stations have buffered unicast data frames waiting for them at the access point.

A DTIM (Delivery Traffic Indication Message) is a special type of beacon frame that informs stations about multicast or broadcast data awaiting delivery.

6) How does a handover work in 802.11? Why is it so slow, compared to GSM or UMTS?

In 802.11 networks, handover involves switching a mobile device's connection from one access point (AP) to another as the device moves within the network coverage area.

It can be slow because of:

- Scan time
- Authentication and association
- Reconfiguration Delay
- Link layer handover

Physical Layer Variants and Extensions:

1) What are the advantages and disadvantages of using higher frequencies?

Advantages:

Increased bandwidth, shorter wavelength, reduced interference, directional antennas

Disadvantages:

Limited range , Penetration loss, Equipment cost , Regulatory restrictions

2) Why is the transmit power limited to 100 mW?

Because of following reasons:

Prevention of interference, mitigation of health risks, spectrum management